

Element H  
Raja Swaminathan

We will test our designs through the following ways:

Time:

1. We will run a simple code snippet that asks for the time at a given moment and see if it is accurate to the current time

Motor:

1. Connect servo tester to motors
2. Connect motor to ring gear and small gear
3. Spin servo tester and thus motor
4. Observe speed and movement of the servo tester

Launcher:

1. We will work with the launching team in order to figure out the range of their launcher
  - a. Insert a marker eraser into the launching mechanism
  - b. Shoot both the crossbow mechanism and air pump mechanism individually
  - c. Observe the speed and launch of the eraser marker

With these tests, we will be able to find out the limitations of our mechanisms, and we can work with our launcher team in order to optimize the mechanisms to suit the zoo's criteria

These tests will all be standardized in the sense that we will run 5 tests of each experiment and we are looking for at least 3 of each trials to be optimal or more than optimal.

```
void loop()
{
```

```
    DateTime now = rtc1.now();
    now.dayOfTheWeek();
    if((now.hour() == goalTime[0]) && (now.minute() == goalTime[1]) && (now.second() ==
30)){
        //run the special code for launch
        lcd.display("Team Geppetto's Automation!");
    }
}
```

The following code displays how we will test the time code and see if it works.